

Cardiff School of Technologies

Assessment Brief

Module Code

Module Title

DAS5003

Blockchain Fundamentals

Academic Year

Semester

2026

1

Module Leader email

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Content

Assessment Details	2
Submission Details	3
Assessment Criteria	3
Further Information	5
Who can answer questions about my assessment?	5
Referencing	5
Submission problems	5
Unfair academic practice	5
How is my work graded?	6

Assessment Details

Assessment title	Abr.	Weighting
A comprehensive report showcasing the understanding of the fundamentals of Blockchain Technology	PRAC1	80%
Pass marks are 40% for undergraduate work and 50% for postgraduate work unless stated otherwise.		

Task/assessment brief:

PRAC1

Section 2 – Blockchain Data Analysis (1000 words) (35%)

For this task, you must use blockchain.com data api. Please familiarise yourself with the api endpoints and read through the documentation so you can identify the correct endpoints to call, to answer the below questions.

For this task, you must use blockchain explorer APIs such as:

- [Blockchain.com API Documentation](#)
- [BlockCypher API](#)
- [Etherscan API](#)

You are required to analyse real-time blockchain data and perform blockchain graph analytics.

Task 2.1 – Real-Time Blockchain Investigation (20%)

Fetch live blockchain data using suitable API endpoints and answer the following questions:

- I. What is the latest mined block height and mining difficulty level? (4%)
- II. Identify the mining address or mining pool associated with the latest block. Can the identity of the miner be inferred or unmasked? (4%)
- III. How many transactions are included in the block? (3%)
- IV. Which wallet address has the highest balance among the analysed addresses? (3%)
- V. Which address has the highest number of transactions? (3%)
- VI. When did this address first become active on the blockchain network? (3%)

Students must include:

- API endpoints used
- JSON/API response screenshots
- Tables and charts summarizing findings
- Brief interpretation of the collected data

Task 2.2 – Historical Blockchain Data Retrieval (15%)

Given the latest block information obtained in Task 2.1, explain how you would retrieve all Bitcoin blockchain transaction data up to 1st January 2023 using blockchain APIs or blockchain explorer services. Your answer should describe the API endpoints or services used, how block heights, timestamps, and pagination can be applied to retrieve historical records, the overall steps required to iteratively collect blocks and transaction data, and the challenges involved such as API rate limits, storage requirements, and large-scale data processing. You may present your explanation using a workflow diagram, algorithm, or pseudocode.

Note: No implementation code is required. A clear explanation of the approach is sufficient.

Maximum word count: 300 words. (15%)

Section 3 (PRAC1) – Blockchain Development (2000 words) (65%)

This task requires you to **work in a team of three**. You will be required to develop a DAPP by drawing from the skills and knowledge gained during the course of this module. As a team, agree on either of the following two use cases (Only develop **one of the two**) based on your interest.

Section 3 – Blockchain Development and Smart Contracts (2000 Words) (65%)

Task 3.1 (a) – Decentralized Digital Asset Marketplace (50%)

Design and develop a blockchain-based decentralized application (Dapp) that supports the creation and management of digital assets.

The application should meet the following requirements:

- Allow users to define and register a digital asset
- Mint blockchain tokens representing ownership of the asset
- Allow users to:
 - purchase,
 - transfer,
 - and sell digital assets
- Maintain ownership history on the blockchain
- Display:
 - total assets created,
 - total transactions,
 - top 10 asset holders,
 - and asset ownership records
- Use smart contracts to automate transactions and ownership transfers
- Integrate wallet support such as MetaMask

Additionally, students should discuss:

- smart contract security,
- gas fees,
- scalability,
- and decentralized trust mechanisms.

Task 3.1 (b) – Decentralized Voting and Governance System (50%)

Design and implement a blockchain-based decentralized voting application.

The application should meet the following requirements:

- Allow authorized users to create elections or voting campaigns
- Allow verified blockchain addresses to vote
- Prevent duplicate voting from the same address
- Allow voters to modify their vote before the deadline
- Record all voting activity immutably on-chain
- Automatically calculate and display election results
- Provide transparency and auditability of votes
- Use smart contracts to manage election logic and voting rules

Task 3.2 – Individual Reflection and Contribution (15%)

- Each student must individually provide a brief reflection describing their overall experience of working on the blockchain project and summarise their specific contribution to the development process. The reflection should include the technical tasks completed, the challenges or bottlenecks encountered during the project, how those challenges were resolved, and the skills or knowledge gained throughout the development process. Students should also discuss their experience of teamwork, collaboration, and the practical understanding they developed regarding blockchain technologies, smart contracts, decentralised applications (DApps), and blockchain-based system development.

Word count (or equivalent):

3200 words equivalent

This is a reflection of the effort required for the assessment. Word counts will normally include any text, tables, calculations, figures, subtitles and citations. Reference lists and contents of appendices are excluded from the word count. Contents of appendices are not usually considered when determining your final assessment grade.

Academic or technical terms explained:

DLT: Distributed Ledger Technologies such as Blockchain, Directed Acyclic Graphs, Block Dags and Hash graphs.

DAPP: Decentralized Application running on the blockchain network

Artificial Intelligence Models – Guidance for this assessment:

Artificial Intelligence (AI) models can be a powerful tool to support your learning. The University has provided some resources to support you in its appropriate usage:

- [Library Services AI Hub](#)
- [Student Guide to AI and Assessment](#)
- [Code of Conduct for Students on the use of AI](#)
- [Cite Them Right resource on citing materials relating to AI \(if permitted\)](#)

As per the academic regulations ([Academic Handbook Ah1 08](#)), in all cases you must submit work that is your own, acknowledging any part of it that has been informed by another source – including that which is AI generated. Upon submission of work, you will be asked to confirm the following statement:

I confirm that this assignment is my own work, except where I have acknowledged the use of works from other sources, including the use of any artificial intelligence (AI) tools, in accordance with what is allowable as described in the assessment brief.

Please note the following:

- AI should not be used as a substitute for your own knowledge, and you should never include any material that you do not understand and could not explain if asked.

- Not being able to explain your work when asked is likely to be a key factor when considering cases of academic misconduct related to AI.

The following information provides specific guidance for this assessment about what level of AI use is appropriate for this assessment. Remember that in all cases you must submit work that is your own, acknowledging any part of it that has been provided by another source.

NO USE OF GENERATIVE AI EXPECTED • Your assignment should be produced using information sourced by you from your learning materials and academic sources and cited appropriately. • AI tools for checking spelling, grammar and referencing may be used.	
AI ACKNOWLEDGED • You can use AI tools to learn about your topic, as part of your study, or in preparing initial guidance on assignments (e.g. headline structure, suggestions for inclusion of topics). • Any materials that you have sourced from AI should be rewritten or reconfigured and integrated into your own work and referenced appropriately. It is recommended that this is confirmed by a relevant academic source. • Any support gained from AI should be acknowledged in a statement at the end of the assignment, making clear what the support was, and how you used it and developed it for your own work. Example statements are available in the Student Code of Conduct [link].	
AI EMBEDDED • Use of AI is an integral and expected part of the assessment. • The explicit inclusion of AI within the assessment means that instructions on the expected use will be part of the assessment brief. • Your assessment brief will describe how you should acknowledge the way in which you used AI tools.	

Submission Details

Submission Deadline:	Check Moodle for submission dates	Estimated Feedback Return Date	This will normally be 20 working days after initial submission.
Submission Time:	By 4.00pm on the deadline day.		
Moodle/Turnitin:	Any assessments submitted after the deadline will not be marked and will be recorded as a non-attempt unless you have had an extension request agreed or have approved mitigating circumstances. See the School Moodle pages for more information on extensions and mitigating circumstances.		
File Format:	The assessment must be submitted as a pdf document (save the document as a pdf in your software) and submit through the Turnitin submission point in Moodle.		
	Your assessment should be titled with your:		

Feedback

Feedback for the assessment will be provided electronically via Moodle. Feedback will be provided with comments on your strengths and the areas which you can improve. View the [guidance](#) on how to access your feedback.

All marks are provisional and are subject to [quality assurance processes](#) and confirmation at the programme Examination Board.

Assessment Criteria

Learning outcomes assessed

Learning Outcomes	LO Assessed
1. Explore essential concepts of blockchain and related distributed ledger technologies.	Section 2
2. Articulate how blockchain can ensure trust, and the difference between centralised and decentralised trust.	Section 3
3. Apply and create crypto currencies, mining, digital assets and value preservation without the middleman.	Section 2 and Section 3
4. Blockchain graph analysis and embedded learning with data visualisation and prediction.	Section 2
5. Explore different types of blockchains, cryptographic algorithms, Distributed Applications (DApp) and decentralized peer-to-peer networks.	Section 2 and Section 3
6. Apply programming techniques to create smart contracts.	Section 3
7. Critically evaluate and identify emerging technologies and research areas relevant to blockchain.	Section 3

Other skills/attributes developed

This includes elements of the Cardiff Met EDGE (Ethical, Digital, Global and Entrepreneurial skills) and other attributes developed in students through the completion of the module and assessment. These will also be highlighted in the module guidance, which should be read by all students completing the module. Assessments are not just a way of auditing student knowledge. They are a process which provides additional learning and development through the preparation for and completion of the assessment.

Ethical	Ethical issues regarding the usage of geospatial and social media data are considered.
Digital	The assignment deliverable will consist of executable Python code for digital representation

	of geospatial data.
Global	The assignment is heavily based on the World Bank data. In addition, global issues like cryptocurrency are considered.
Entrepreneurial	The solution to part 2 involves applications of geospatial data in gauging public sentiment which could be translated into entrepreneurial activities such as market research, think tanks etc.

Marking/Assessment Criteria

Assessment Criteria	100%
Task 2.1 (LO1, LO2, LO3, LO4)	30%
Task 2.2 (LO1, LO4)	5%
Task 3.1 (LO2, LO5, LO6, LO7)	50%
Task 3.2 (LO3, LO6, LO7)	15%

70 – 100% (1st)	Addressed all sections and provided correct answers with elegant presentation of results. Applied correct data analysis approaches and provided excellent interpretation on each section.
60-69% (2:1)	Addressed all sections and provided correct answers with good presentation of results. Applied mostly correct data analysis approaches and provided very good interpretation on each section.
50-59% (2:2)	Addressed most of the sections and provided mostly correct answers with average presentation of results. Applied some correct data analysis approaches and provided an average interpretation on each section.
40-49% (3rd)	Addressed few sections with few correct answers with/out any presentation of results. Applied mostly incorrect data analysis approaches and provided poor interpretation on each section.
35-39% (Narrow Fail)	Addressed few sections and provided mostly incorrect answer with poor presentation of results. Applied incorrect data analysis approaches and provided poor interpretation.
<35% (Fail)	Very poor report missing one or more required parts.

Further Information on assessment, referencing and grading can be found in the Module Handbook (on Moodle)

Further Information

Who can answer questions about my assessment?

Questions about the assessment should be directed to the staff member who has set the task/assessment brief. This will usually be the Module Leader. They will be happy to answer any queries you have.

Staff members can often provide feedback on an assignment plan but cannot review any drafts of your work prior to submission. The only exception to this rule is for Dissertation Supervisors to provide feedback on a draft of your dissertation.

Referencing and independent learning

Please ensure you reference a range of credible sources, with due attention to the academic literature in the area. The time spent on research and reading from good quality sources will be reflected in the quality of your submitted work.

Remember that what you get out of university depends on what you put in. Your teaching sessions typically represent between 10% and 30% of the time you are expected to study for your degree. A 20-credit module represents 200 hours of study time. The rest of your time should be taken up by self-directed study.

Unless stated otherwise you must use the **HARVARD** referencing system. Further guidance on referencing can be found in the Study Smart area on Moodle and at www.citethemrightonline.com (use your university login details to access the site). Correct referencing is an easy way to improve your marks and essential in achieving higher grades on most assessments.

Technical submission problems

It is strongly advised that you submit your work at least 24 hours before the deadline to allow time to resolve any last minute problems you might have. If you are having issues with IT or Turnitin you should contact the IT Helpdesk on (+44) 2920 417000. You may require evidence of the Helpdesk call if you are trying to demonstrate that a fault with Moodle or Turnitin was the cause of a late submission.

Extensions and mitigating circumstances

Short extensions on assessment deadlines can be requested in specific circumstances. If you are encountering particular hardship which has been affecting your studies, then you may be able to apply for mitigating circumstances. This can give the teachers on your programme more scope to adapt the assessment requirements to support your needs. Extensions and mitigating circumstances policies and procedures are regularly updated. You should refer to your degree programme or school Moodle pages for information on extensions and mitigating circumstances.

Unfair academic practice

Cardiff Met takes issues of unfair practice **extremely seriously**. The University has procedures and penalties for dealing with unfair academic practice. These are explained in full in the University's Unfair Practice regulations and procedures under [Volume 1, Section 8](#) of the Academic Handbook. The Module Leader reserves the right to interview students regarding any aspect of their work submitted for assessment.

Types of Unfair Practice, include:

Plagiarism, which can be defined as using without acknowledgement another person's

words or ideas and submitting them for assessment as though it were one's own work, for instance by copying, translating from one language to another or unacknowledged paraphrasing. Further examples include:

- Use of any quotation(s) from the published or unpublished work of other persons, whether published in textbooks, articles, the Web, or in any other format, where quotations have not been clearly identified as such by being placed in quotation marks and acknowledged.
- Use of another person's words or ideas that have been slightly changed or paraphrased to make it look different from the original.
- Summarising another person's ideas, judgments, diagrams, figures, or computer programmes without reference to that person in the text and the source in a bibliography/reference list.
- Use of assessment writing services, essay banks and/or any other similar agencies (NB. Students are commonly being blackmailed after using essay mills).
- Use of unacknowledged material downloaded from the Internet.
- Re-use of one's own material except as authorised by your degree programme.

Collusion, which can be defined as when work that has been undertaken with others is submitted and passed off as solely the work of one person. Modules will clearly identify where joint preparation and joint submission are permitted, in all other cases they are not.

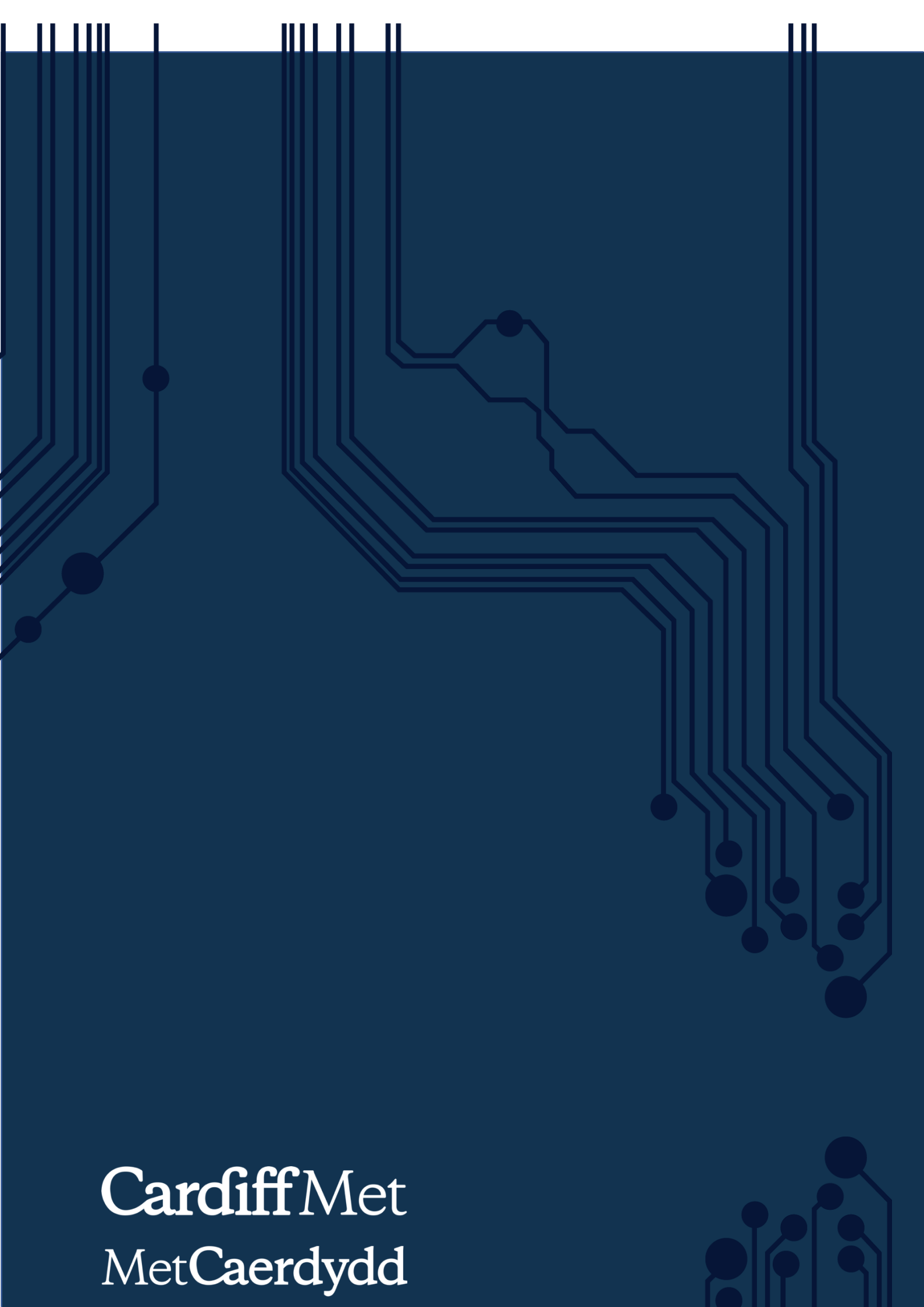
Fabrication of data, making false claims to have carried out experiments, observations, interviews or other forms of data collection and analysis, or acting dishonestly in any other way.

How is my work graded?

Assessment grading is subject to thorough quality control processes. You can view a summary of these processes on the [Assessment Explained Infographic](#).

Grading of work at each level of Cardiff Met degree courses is benchmarked against a set of general requirements set out in [Volume 1, Section 4](#) of our Academic Handbook. A simplified version of these Grade Band Descriptors (GBDs) with short videos explaining some of the academic terminology used can be accessed for [Foundation](#), [1st year](#), [2nd year](#) and [3rd year](#) undergraduate and [MSc programmes](#).

We would strongly recommend looking at the [Study Smart](#) area of Moodle to find out more about assessments and key academic skills which can have a significant impact on your grades. Always check your work thoroughly before submission

An abstract graphic design featuring a dark blue background with a complex, light blue circuit-like pattern. The pattern consists of numerous thin, parallel lines that branch out and connect to various circular nodes of different sizes. The lines and nodes are arranged in a way that suggests a network or a data flow, with some lines running vertically and others branching out horizontally or diagonally. The overall effect is a modern, technological aesthetic.

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